BRIEF GLOSSARY

1. **Ecology in the Anthropocene**
   a. **Ecology** - the science of how organisms (including humans) interact with each other and with their physical environment; it asks why organisms are distributed and abundant as they are, and how biotic and abiotic processes shape those patterns.
   b. **Evolution** - change in the heritable characteristics (allele frequencies, traits) of populations across generations, driven by processes such as mutation, natural selection, genetic drift, migration, and recombination.
   c. **Natural selection** - a mechanism of evolution in which heritable traits that increase survival or reproduction become more common over generations because their carriers leave more offspring than alternative variants.
   d. **Speciation** - the process by which new species arise, usually when populations become genetically differentiated and reproductively isolated so that gene flow between them is strongly reduced or absent.
   e. **Extinction** - the process by which a species (or lineage) disappears when its last individuals die or fail to reproduce; unlike local extirpation, true extinction means that no viable populations remain anywhere.
   f. **Habitat** - the physical and biotic environment where an organism lives, including climate, substrate/vegetation, resources, and other species.
   g. **Niche** - the multidimensional "role and requirements" of a species: the range of abiotic conditions and resources it can use and the biotic interactions it engages in. The fundamental niche is set by physiology and resources; the realized niche is what it actually occupies in the presence of competitors, predators, and other constraints.
2. **Eco-evolutionary dynamics & life histories**
   a. **Life-history strategies/traits** - characteristics of an organism related to key organismal traits and events (growth, reproduction, survival, age at maturity, number and size of offspring, lifespan) and how those are shaped by environments.
   b. **Slow–fast continuum** - a major axis of life-history variation ranging from "fast" strategies (early reproduction, many small offspring, short lifespan, high mortality) to "slow" strategies (late reproduction, few large offspring, long lifespan, low mortality).
   c. **Pace-of-life syndromes** - correlated suites of life-history, physiological, and behavioural traits along a slow–fast continuum (e.g., fast-growing, bold, high-metabolism individuals vs. slow-growing, shy, low-metabolism individuals), reflecting consistent differences in how organisms allocate energy and risk through time.
3. **Population ecology essentials**
   a. **Allee effect (component vs demographic)** - The component Allee effect is the positive relationship between any measurable component of individual fitness and population density. The demographic Allee effect is the positive relationship between the overall individual fitness and population density.
   b. **Allee effect (strong vs weak)** - a strong Allee effect produces a critical threshold density below which the population's per-capita growth rate is

negative and extinction is likely; a weak Allee effect increases growth rate with density at low densities but does not create a hard threshold - the population can still, in principle, recover from very low numbers.

   c. **Rescue/immigration effects** - the prevention or delay of local extinction by the arrival of immigrants from other populations.

4. **Species interactions & coexistence**
   a. **Predation** - a consumer–resource interaction that reduces a prey's future survival and/or reproduction to 0, not just "kill-and-eat." This spans lethal predation (ambush/pursuit), parasitoids that inevitably kill hosts, parasitic castration that eliminates host fecundity, plus herbivory/seed predation that removes future offspring.
   b. **Top-down control** - occurs when consumers (predators, herbivores) primarily regulate the abundance and dynamics of lower trophic levels.
   c. **Bottom-up control** - occurs when resource supply (nutrients, primary production) primarily determines the abundance and dynamics of higher trophic levels.
   d. **Trophic cascades** - chains of direct & indirect effects propagating through food webs when a change at one trophic level (e.g., predators) alters the abundance or behaviour of organisms at lower levels (e.g., herbivores), leading to alternating increases and decreases in biomass or function across multiple trophic levels (e.g., plants).
5. **Food webs & network ecology**
   a. **Keystone species** - species that have a disproportionately large effect on community structure or ecosystem function relative to their biomass or abundance; removing a keystone can trigger large shifts in species composition, trophic cascades, or even regime shifts.
   b. **Interaction strength** - a measure of how strongly one species' presence, abundance or activities (e.g., behaviour) affects another species' vital rates, abundance or activities (e.g., per-capita effect of a predator on prey mortality or of a mutualist on partner reproduction).
   c. **Network structure (degree, modularity, motifs)** - describes how species and their interactions are arranged in a food web or interaction network. Some species have many links (high degree) and can act as hubs; others form semi-independent clusters (modules) that interact more within than between clusters; and certain small link patterns (motifs) recur more often than expected by chance. These structural features strongly influence how robust the web is to species loss and how disturbances spread through the community.
   d. **Network rewiring** - changes in who interacts with whom over time, as existing links weaken or disappear and new links are formed. Rewiring can be driven by natural variation (e.g. seasons, species' range shifts, evolution) or by human activities (e.g. land-use change, species introductions, climate change). It can sometimes buffer ecosystems by providing alternative interaction pathways, but it can also spread disturbances or create novel, risky links (e.g. new predator–prey or host–pathogen interactions).
6. **Macroecology & biogeography**
   a. **Latitudinal diversity gradient** - species richness tends to rise from poles to equator; this is a global, general pattern across most taxonomic groups (e.g.,

vertebrates, invertebrates, and plants); mechanisms include differences in speciation, extinction, and dispersal.

b. **Evolutionary rates & diversification -** evolutionary rate is the speed at which traits or lineages change over time (e.g., morphological evolution per million years); diversification is the net outcome of speciation minus extinction. Together, they describe how quickly lineages originate and accumulate, shaping large-scale patterns of diversity such as the latitudinal gradient.
c. **Taxonomic diversity -** captures how many species (or higher taxa) are present in a community and how evenly individuals are distributed among them.
d. **Functional diversity -** describes the range and distribution of species' traits and ecological roles (e.g., diet, body size, dispersal mode) in a community.
e. **Phylogenetic diversity -** measures how much evolutionary history is represented in a community, typically estimated by the total branch length or relatedness among co-occurring species.
f. **Species concept -** a framework for defining what counts as a "species," such as the biological species concept (groups of actually or potentially interbreeding populations reproductively isolated from others), morphological species concept (based on distinct phenotypes), or phylogenetic species concept (smallest diagnosable monophyletic lineages). Different concepts highlight different processes and can produce different species counts.
g. **Cryptic diversity -** the presence of multiple genetically distinct, reproductively isolated lineages that are lumped under a single nominal species because they are morphologically very similar; cryptic species inflate real diversity relative to current taxonomy.
h. **Extrapolation (in richness estimation) -** the use of observed data to statistically estimate total species richness beyond the sampled range, often via species–accumulation models or non-parametric estimators that account for unseen rare species.
i. **Ecogeographical rules -** empirical patterns relating organismal traits (e.g., body size, shape, or coloration) to broad-scale environmental gradients, such as Bergmann's rule (larger bodies in colder climates), Allen's rule (shorter extremities in cold regions), Gloger's rule (darker pigmentation in warm, humid areas) or island rule (small-bodied species tend to evolve larger body sizes and large-bodied species tend to evolve smaller body sizes on islands compared to mainland relatives).
j. **Insular dwarfism** - large animals evolving or having a reduced body size on islands, typically linked to limited resources and relaxed predation, which favor smaller, more resource-efficient individuals.
k. **Insular gigantism -** small animals evolving or having an increased body size on islands, presumably associated with reduced predation and competition, opening ecological opportunities for larger, more competitive individuals.
l. **Meta-analysis** - quantitative syntheses that statistically combine effect sizes or results across many independent studies to estimate overall patterns, test the robustness of hypotheses, and explore how effects vary with moderators (e.g., taxa, methods, environments).

7. **Global change ecology**
   a. **Background vs contemporary extinction rates -** background rates are the typical long-term extinction rates inferred from the fossil record during

geologically “normal” periods; contemporary rates are the current, often much higher, rates of species loss driven by human activities (habitat change, exploitation, climate change, invasions, pollution), allowing comparisons that diagnose whether we are in a mass-extinction-like episode.

   b. **Extinction debt -** the future loss of species already “committed to extinction” due to past habitat destruction, fragmentation, or other pressures; populations may persist for decades or centuries before finally disappearing, creating a time lag between drivers and realized extinctions.
   c. **“Big Five” mass extinctions & sixth mass extinction -** the Big Five are five geologically recognized periods when ≥75% of species went extinct in relatively short periods (End-Ordovician, Late Devonian, End-Permian, End-Triassic, End-Cretaceous). The proposed sixth mass extinction refers to the idea that current human-driven losses may reach comparable magnitudes, though its status and metrics are actively debated.

8. **Drivers of biodiversity crisis**
   a. **Biodiversity crisis -** global declines in species, populations, and ecosystem integrity driven by anthropogenic habitat change, exploitation, climate, pollution, and invasions.
   b. **Direct vs indirect drivers -** direct drivers are the measurable physical, biological, or chemical pressures that immediately change ecosystems (e.g. habitat conversion, harvesting, pollution, introductions, greenhouse-gas emissions), whereas indirect drivers are the social, economic, demographic, technological, and institutional processes that modify those direct pressures in space and time.
   c. **Biomass reallocation (humans/livestock) -** the large shift of global vertebrate biomass from wild species to humans and their domesticated animals (especially cattle and pigs), such that wild terrestrial mammals now represent only a small fraction of total mammal biomass, with major implications for ecosystem functioning and conservation.

9. Biodiversity ↔ ecosystem functioning/services
   a. **Ecosystem services -** Nature provides regulating (e.g., climate regulation, flood protection, water purification), material (e.g., crops, fish, timber, medicinal plants), and non-material (e.g., outdoor recreation, aesthetic enjoyment, cultural identity) benefits to humans.
   b. **Intrinsic value -** the idea that nature (species, ecosystems) has worth in and of itself, independent of human uses, preferences or economic value.
   c. **Instrumental value -** the usefulness of nature for achieving human needs (e.g., food, water purification, climate regulation).
   d. **Relational value -** the value people see in nature because of their *relationships* with it: feeling at home in a familiar landscape, caring for a forest as a steward, or seeing a species as part of family or community tradition, beyond pure utility or inherent worth.
   e. **Nature-based solutions -** actions that conserve, restore, or sustainably manage ecosystems to address societal challenges (climate mitigation/adaptation, disaster risk reduction, food and water security) while simultaneously providing biodiversity and human well-being benefits.

10. **Applied ecology: new tools & data streams**

a. **Citizen science (GBIF/eBird)** - citizen science = members of the public contribute observations or checks that feed real research. One of examples is eBird (https://ebird.org/) that collects bird observations with explicit effort (e.g., start time, duration). GBIF (https://www.gbif.org/) aggregates occurrence records and higher-quality sampling-event datasets that include methods and effort metadata. Because these data are semi-structured and biased toward easy places/times, analyses should (i) filter for quality (e.g., eBird complete checklists with bounded effort; sampling-event over bare occurrences) and (ii) model the observation process (include effort covariates or use occupancy–detection frameworks).
b. **Ten Principles of Citizen Science** - a widely used set of guidelines that define good practice in citizen science, covering aspects such as genuine participation in scientific activities, clear communication of goals and outcomes, attention to data quality and ethics, appropriate recognition of volunteers, and open sharing of results.
c. **Participation levels (e.g., Haklay typology)** - a classification of citizen-science projects based on how deeply volunteers are involved, from crowdsourcing (volunteers as data collectors), through projects where volunteers also help sort and interpret data, to participatory science and extreme citizen science, where participants help co-design questions, methods, analysis, and application of results.